Means and Variances of Linear Combinations of Random Variables along with Covariance

1. X and Y are independent random variables with means 3, 2 and the variances 1, 4 respectively. The random variable Z is defined as

$$Z = 3X + Y$$

(a) Evaluate Cov$(X, 2Z)$

2. The random variables X, Y and Z have means $\mu_X = 3$, $\mu_Y = -4$ and $\mu_Z = 2$, the variance $\sigma_X^2 = 2$, $\sigma_Y^2 = 1$ and $\sigma_Z^2 = 5$ and $Cov(X, Y) = -1$, $Cov(X, Z) = 1$ and $Cov(Y, Z) = -2$

(a) Find the mean and variance of the random variable:

$$W = 3X - 2Y + Z$$

3. The random variables X, Y and Z have means $\mu_X = 1$, $\mu_Y = -1$ and $\mu_Z = 3$, the variance $\sigma_X^2 = 4$, $\sigma_X^2 = 2$ and $\sigma_Z^2 = 3$ and $Cov(X, Y) = -1$, $Cov(X, Z) = 0$ and $Cov(Y, Z) = -2$

(a) Find the mean and variance of the random variable:

$$U = X - Y + Z$$

(b) Evaluate the covariance Cov$(2X, 3U)$

4. The random variables X, Y and Z have means $\mu_X = 1$, $\mu_Y = -3$ and $\mu_Z = 4$, the variance $\sigma_X^2 = 1$, $\sigma_X^2 = 5$ and $\sigma_Z^2 = 2$ and $Cov(X, Y) = -2$, $Cov(X, Z) = -1$ and $Cov(Y, Z) = 1$

(a) Find the mean and variance of the random variable:

$$W = 3X - Y + 2Z$$